



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Summer 2025

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 40 Duration: 2 hours

Answer all questions. Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Read the following code and detect different types of errors. 8

```
1. include<Math.h>
2. int strLen (char x[], int n){
3. int len = -1;
4. while(char x[len]!=n)
5.     len++;
6. return lens;

7. int main(){
8. int a1 = 0, b1 = 1,,c2=0, 1a;
9. char str[5] = "Hello";
10. a1 = b1 + c;
11. temp = strLen(str, 10);
12. a1 = a1>b1 ? b1 : a1 ;
13. printf("%f is greater\n"a1)
14. if(b1/c2==1)
15.     strlen(x, c2));
16. printf("Not Done : %d", sqrt(a1,2));
17. return c2;
18.}
```

2. Generate Recursive Predictive Parsing Table for the following grammar with FIRST and FOLLOW sets. [3]

```
C → 18c | [A] | ∈
G → +P25BA | -
A → 1G | A1cP | P1G
P → CB75A
B → /Cc111 | +P* | A
```

3. Construct a Non-recursive Predictive Parsing table for the following grammar and check whether string “**caddbc**” is accepted or not. [3]

```
A → BcA1 | cA1
A1 -> aA1 | ∈
B → AdB1 | dB1
B1 → bB1 | ∈
```

4. Eliminate immediate left recursion from the following grammar. [3+3]
- (a) $E \rightarrow E+E \mid (E) \mid EA \mid *A$
 $A \rightarrow AB \mid AAC \mid Ba \mid Cb$
 $B \rightarrow B + B \mid BBAD \mid B(BA) \mid b \mid \epsilon$
 $D \rightarrow DC \mid a/C \mid d/C$
 $C \rightarrow (C) \mid a \mid b \mid c \mid E$
- b) $E \rightarrow EABx \mid Eb \mid 6 \mid \epsilon \mid aEEb \mid EaEE \mid abc \mid BaE$
 $A \rightarrow c \mid A[C] \mid AA*B9D+ \mid AA \mid aD \mid ABC$
 $B \rightarrow A+EBa \mid [AEa] \mid bA \mid 0B0 \mid BB10 \mid B1B10$
 $C \rightarrow AED \mid abc \mid xCy$
5. Left Factor the following. [4+4]
- (a) $X \rightarrow \text{Poet Loves Poetry} \mid \text{Poets Loves eat Sugar} \mid \text{Parrot talks} \mid \text{palm tree} \mid \text{Panache} \mid \text{Poker face} \mid \text{polite nature} \mid \text{polishing} \mid \text{Poetic} \mid \text{police chase} \mid \text{polish} \mid \text{parochial} \mid \text{paradox} \mid \text{persisting}$
- b) $Y \rightarrow [+*[*3)2) \mid /*754 \mid /*po[o[s \mid p[abc] \mid [/*23 \mid $*^{\wedge}\& \mid 8)[a_ \mid /[*231 \mid [+*[*23\# \mid 8) \mid (+)2) \mid p[s[s] \mid /*po(t) \mid \epsilon \mid 8)[b_c]$
6. For the following grammars, write down the FIRST and FOLLOW sets for each nonterminal. [4+4+4]
 Consider all the terminals are of single character/number/symbol.
- (a) $X \rightarrow eXYT-h \mid iXPrq \mid YtPrq \mid Pgh \mid cd$
 $Y \rightarrow pqX*P \mid c \mid d \mid Piy \mid TroXg \mid \epsilon$
 $P \rightarrow Yq \mid PcTXp \mid iqXfYqXY/ \mid cY \mid \epsilon$
 $T \rightarrow e \mid u \mid YP+ \mid iYrTP \mid \epsilon$
- (b) $X \rightarrow ZYX \mid a \mid cYrc$
 $Y \rightarrow XZ* \mid bc \mid XY+Wt \mid \epsilon$
 $Z \rightarrow a \mid WYR$
 $R \rightarrow bob \mid W \mid \epsilon$
 $W \rightarrow johnZ \mid bRaW/ \mid \epsilon \mid x$
- (c) $Z \rightarrow YHXXH \mid SH6aVoWi$
 $Y \rightarrow V*HahaW+ \mid \epsilon$
 $X \rightarrow x \mid \epsilon$
 $W \rightarrow laWY-gH \mid XHZnV!nXi$
 $V \rightarrow 13S31 \mid 24W42 \mid 57Z75$
 $S \rightarrow YZngWZdeSa \mid XHakZ$
 $H \rightarrow \epsilon \mid y$